

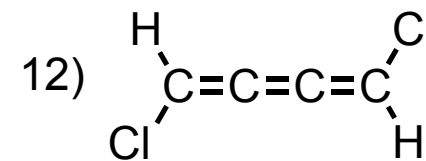
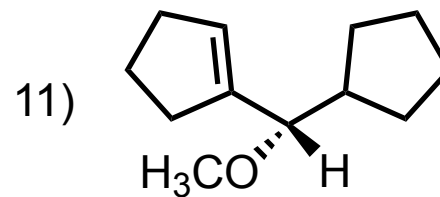
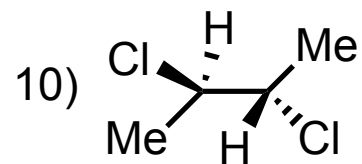
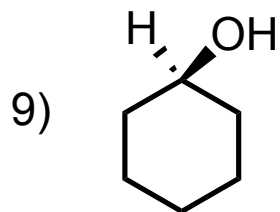
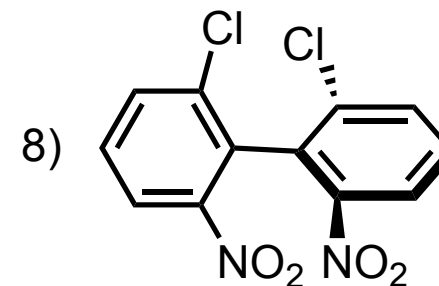
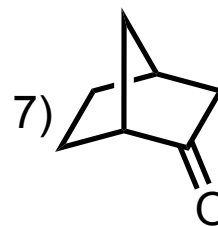
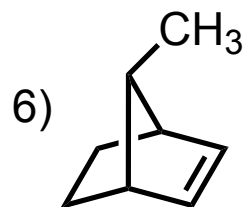
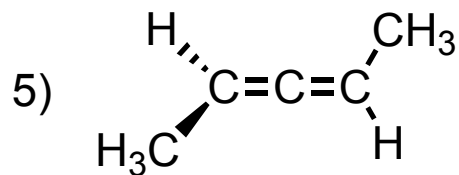
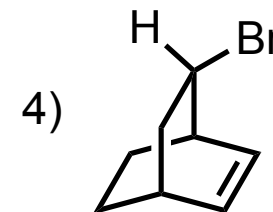
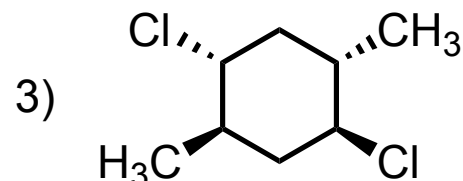
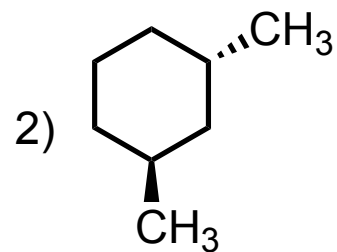
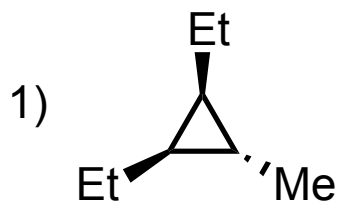
CML-101

Organic Chemistry

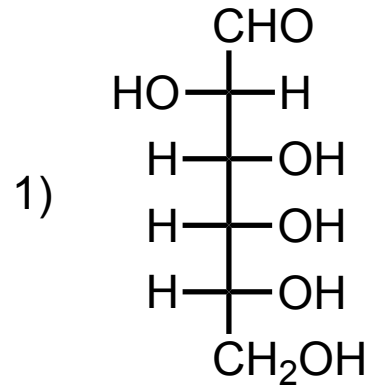
Tutorial-1

Importance of stereochemistry in drug action (warfarin, thalidomide, naproxen); R/S nomenclature, Fisher, Sawhorse, and Newman projections

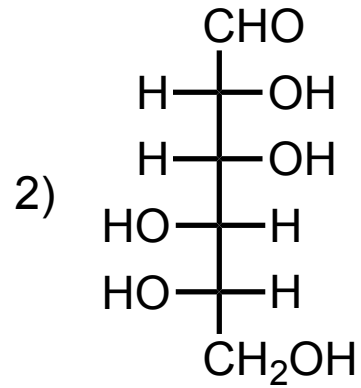
1) Identify each of the following molecule as chiral or achiral?



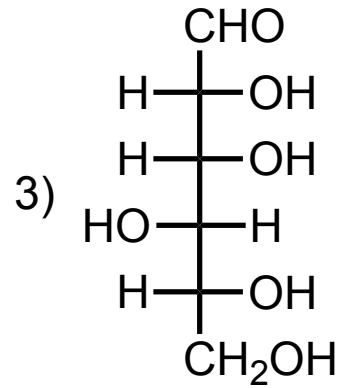
2) Designate D/L configuration for the following molecules.



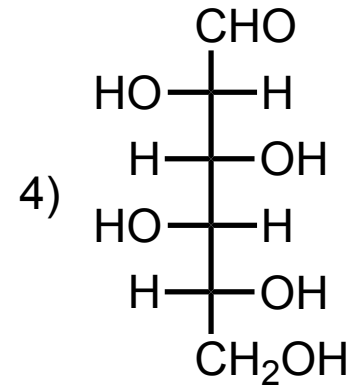
Altrose



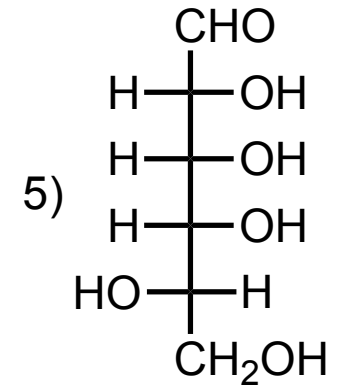
Mannose



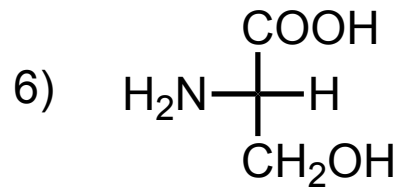
Gulose



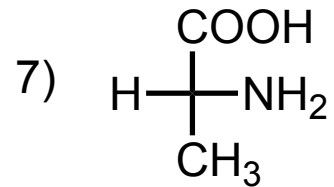
Iodose



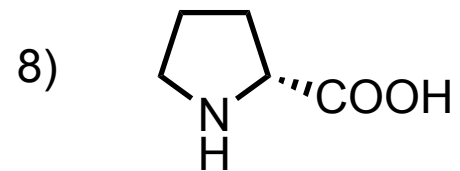
Talose



Serine

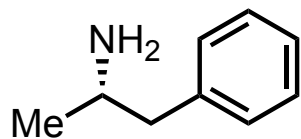


Alanine

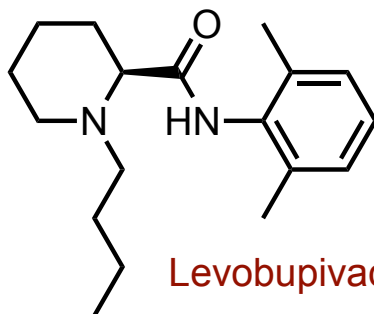


Proline

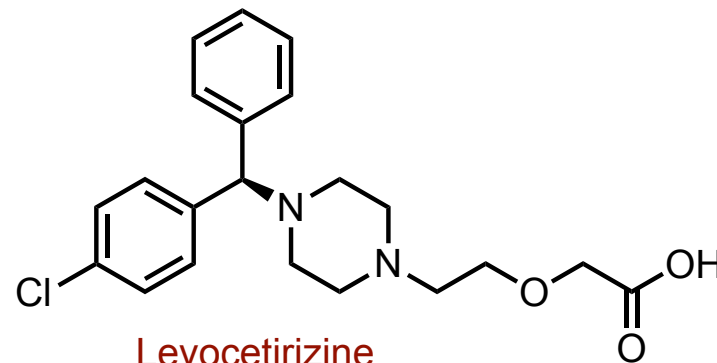
3) Assign R/S configuration for the following commercially selling medicinal compounds.



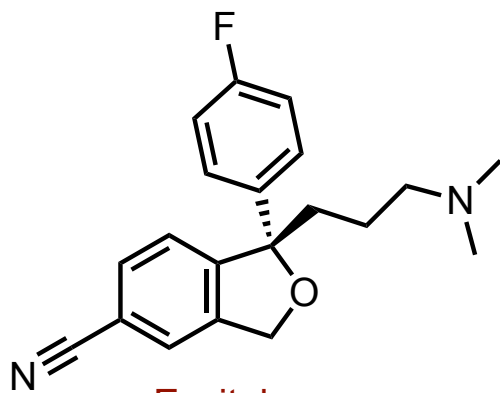
Dextroamphetamine



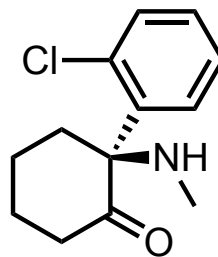
Levobupivacaine



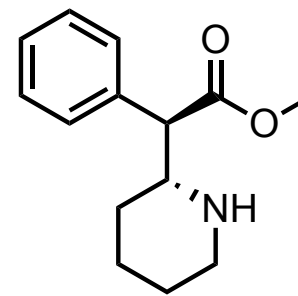
Levocetirizine



Escitalopram

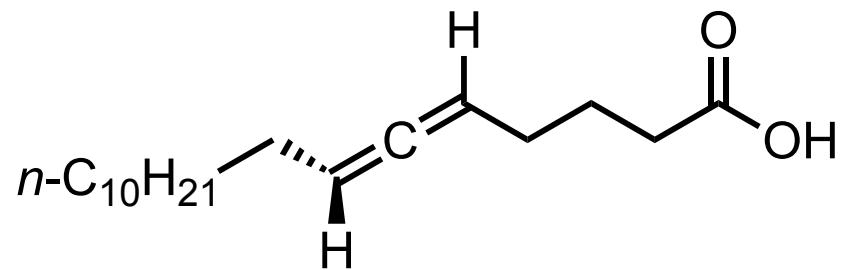


Esketamine

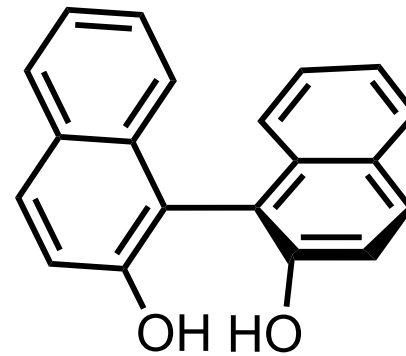


Dexmethylphenidate

4) Assign R/S configuration for the following compounds.

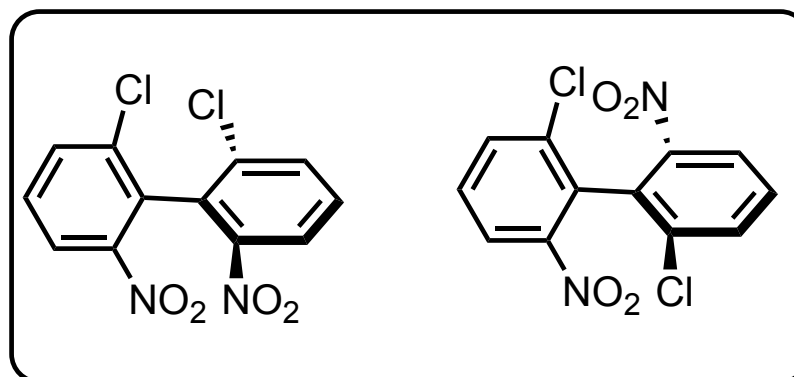
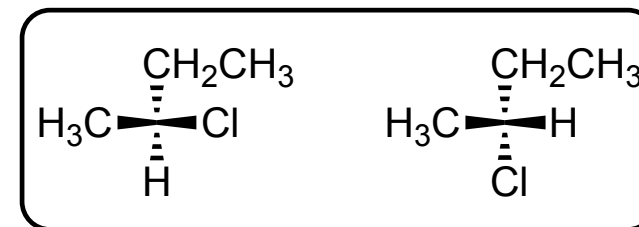
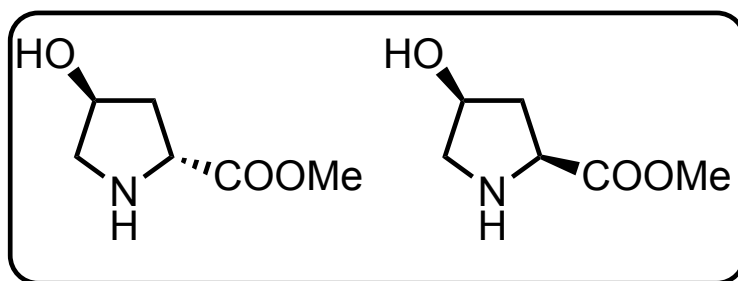
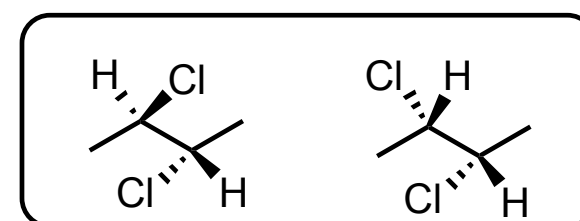
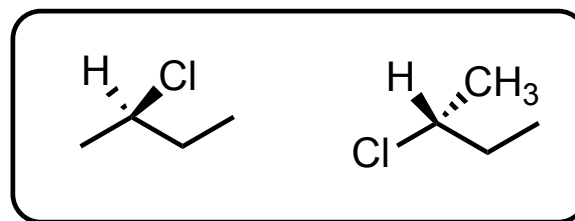
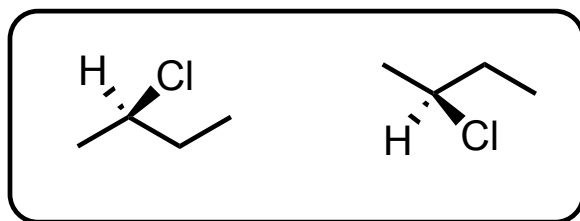


laballenic acid



1,1'-Bi-2-naphthol (BINOL)

5) Identify the relationship between following stereoisomers (Enantiomers/Diastereomers/same molecules).

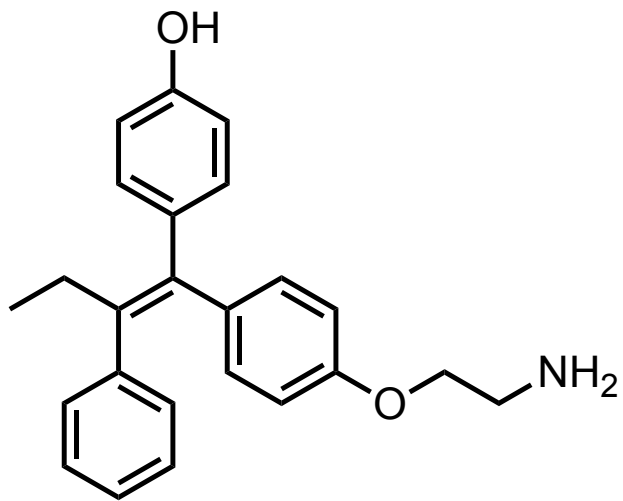


6) Draw the Wedge, Fischer, Newman and Sawhorse projections for the following compounds

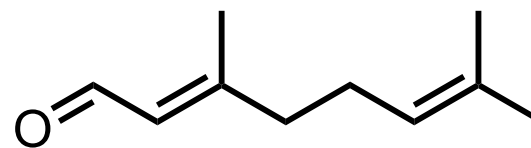
1) (2S,3S)-3-bromobutan-2-amine

2) (2R,3R)-2-bromo-3-fluorobutanoic acid

7) Assign E/Z configuration for the following molecules.



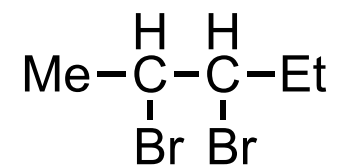
Norendoxifen



citral

8): A sample containing a single enantiomer of fluoxetine (Prozac) is placed in a sample tube (length = 12.5 cm) of polarimeter. The observed rotation is 9.06° clockwise at 25° C . The sample was made by dissolving 1.24 g of fluoxetine in a solution with a total volume of 2.62 mL. Calculate the specific rotation of the sample.

9) Draw the possible number of stereoisomers for the following compound and also indicate the enantiomeric pairs .



10) How can you separate *rac*-2-pentanol using following chiral acid chloride?

